PERMUTATIONS & COMBINATIONS-III

1. How many ways from a group of 5 students 3 are to be chosen for the competition?
2. How many ways from a group of 15 students 6 are to be chosen for a competition?
3. In a party there are 15 persons, how many handshakes are possible if every person and shake with every other person?
4. Out of 6 consonants and 4 vowels how many words of 3 consonant and 2 vowels can be formed?
5. In how many ways can a student choose a program of 5 courses if 9 courses are available and 2 specific courses are compulsory for every student?
6. In how many ways can a group of 5 men and 2 women be selected out of a total of 7 men and 3 women?
7. In how many ways one can select a cricket team of 11 from 17 players in which only 5 players can bowl if each cricket team of 11 players must include exactly 4 bowlers?
8. A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways this can be done when the committee consists of
   (i) Exactly 3 girls (ii) At least 3 girls (iii) At most 3 girls
9. A bag contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the bag if at least one black ball is to be included in the drawn?
10. In an examination paper, there are two groups each containing 4 questions. A candidate is required to attempt 5 questions but not more than 3 questions from any group. In how many ways can 5 questions be selected?
11. A question paper has two parts, A and B, each containing 10 questions. If a student has to choose 8 from part A and 5 from part B, in how many ways can he choose the questions?
12. There are 6 classes in each working day of school. In how many ways can one organize 5 subjects such that each subject is allowed at least one class?

PRACTICE QUESTIONS:

1. In how many ways one can select a badminton team of 2 players out of 4 players?
2. In how many ways one can select a cricket team of 11 players from 17 players?
3. A bag contains 4 black and 4 red balls. Determine the number of ways in which 2 black and 3 red balls can be selected?
4. In how many ways can a team of 3 boys and 3 girls be selected from 5 boys and 4 girls?
5. How many words, with or without meaning, each of 3 vowels and 2 consonants can be formed from the letters of the word INVOLUTE?
6. Determine the number of 5-card combinations out of a deck of 52 cards if each selection of 5 cards has exactly one king?
7. Determine the number of 5 card combinations out of a deck of 52 cards if there is exactly 2 Ace in each combination?
8. In how many different ways can 3 people be seated at a circular table?
9. We have 1 red, 1 blue and 1 purple flower. In how many different ways can we arrange them to make a garland?

CODEGNAN